	Title	PARI Standard High Carbon & Alloy Steel Material Selection wrt. it's Heat Treatment Properties and Application	Document No	PRD/GD/DES/11
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	Classification	Reference - Guideline	Effective Date	09-Mar-16

i. Raw material given in the below table to be selected & specified on drawing by Designer based on it's Heat Treatment Properties and application requirement.

ii. Requirements of Forging / Magnetic Particle testing / Ultrasonic testing / Dye Penetrant testing / Stress Relieving to be specified by Designer on drawing as per application requirement.

iii. Any raw material apart from below table can be selected by Designer for the specific reason of special requirements and specify such raw material with Heat treatment requirements & NDT requirements as applicable on drawing.

Material, Heat Treatment Type and Hardness range

Sr. No.	Material (with Equivalent IS/BS/SAE No Material)	PARI STD	Raw Material Notes	Toughening process Hardness Range (HRC)	Flame Hardening process Hardness Range (HRC) *[Area to be hardened to be shown by XXXXX on drawing].*	Induction Hardening process Hardness Range (HRC) *[Area to be hardened to be shown by XXXXX on drawing].*	Case Hardening (HRC), Case Depth Range (HRC) *[Area to be hardened to be shown by XXXXX on drawing].*	Special Heat Treatment Process and Hardness Range (HRC)	Through Hardening Hardness Range (HRC)	UTS (Kg/mm²)	Allowable Stress (Kg/mm²)	Typical application parts	Additional Information
1	C45 / C55 (En8D/En9)	EN9	Round bar stock of <= 32 dia. (Refer Sheet 2)	24-32 HRC	45 - 55 HRC, 0.6 to 2 mm depth	45 - 55 HRC, 0.6 to 2 mm depth	NIL	Nitriding 45 - 55 HRC, 20 to 25 microns depth	40-45 HRC	60-75	30-37.5	Applicable for parts in Round form like Shafts, keys, fasteners, Wear resistance parts -Pins, shafts	Nitriding to be done after finish machining/ Preferred to be toughened/ machined and then nitrided. (Chrome plating to be avoided because of peeling possibility).
2	C45 / C55	C45	Plate & block stock (Refer Sheet 2)	24-32 HRC	45 - 55 HRC, 0.6 to 2 mm depth	45 - 55 HRC, 0.6 to 2 mm depth	NIL	NIL	40-45 HRC	60-75	30-37.5	Applicable for only parts in Plate form like Wear Plates, Resting pad	Low hardenability (Toughened part at core shows low hardness (10 to 14 HRC).
3	40Ni6Cr4Mo3 (EN24/5517)	EN24	= > 32 dia. (Refer Sheet 2)	24-32 HRC	45 - 55 HRC, 0.6 to 2 mm depth	45 - 55 HRC, 0.6 to 2 mm depth	NIL	NIL	50 - 55 HRC	75-130	37.5-65	Bolts, Gears, Shafts, Good for impact loads. Higher strength than 40Cr4 / EN19	--
4	20MnCr5 (SAE8620)	SAE8620	(Refer Sheet 2)	NIL	NIL	NIL	58 ± 2 HRC, 0.6 to 2 mm depth	NIL	NIL			For applications other than shafts/Gears (like Rollers)	--
5	20MnCr5 (4432, EN353)	20MnCr5	(Refer Sheet 2)	NIL	NIL	NIL	58 ± 2 HRC, 0.6 to 2 mm depth	NIL	NIL	62-74	31-37	For Shafts/Gears, locating pins etc.	--
6	EN19	EN19	(Refer Sheet 2)	NIL	50-55 HRC, 0.6 to 2 mm depth	50-55 HRC, 0.6 to 2 mm depth	NIL	Gas Nitriding 50-55 HRC, 20 to 25 micron case depth	NIL	70-110	35-55	For Rest pad	Nitriding to be done after finish machining/ Preferred to be toughened/ machined and then nitrided (Chrome plating to be avoided because of peeling possibility).
7	EN41B	EN41B	--	NIL	NIL	NIL	NIL	Gas Nitriding 50-55 HRC, 20 to 25 microns case depth	NIL	80-100	40-50	For Tool holders	Nitriding to be done after finish machining/ Preferred to be toughened/ machined and then nitrided (Chrome plating to be avoided because of peeling possibility).
8	XT160Cr12 (4430, HCHCr/ D2)	D2	Consult Purchase for complex or large shapes & sizes	NIL	NIL	NIL	NIL	NIL	60 ±2 HRC			Gauges, Punches, Cam plates	--
9	X04Cr18Ni11 (EN58E/SS304)	SS304		NIL	NIL	NIL	NIL	NIL	NIL			For Food grade and Antirust applications (like Washing m/c tanks/doors)	--
10	80C6 (1074)	EN42		NIL	NIL	NIL	NIL	NIL	NIL			Tension/Compression Springs	--
11	FG260	--		NIL	NIL	NIL	NIL	NIL	NIL			Simple applications like belt pulleys	--
12	FG300	--		NIL	NIL	NIL	NIL	NIL	NIL			Rope Sheaves/Pulleys	--
13	OHNS (O1,O2)	OHNS (O1,O2)		NIL	NIL	NIL	NIL	NIL	58 - 62 HRC			For Gauges Probes	--

Raw Material	Standard Raw Material Sizes Available in the Market	Buffer Stock Maintained in PARI for Sizes
EN9	**Black Round :** 10,16,25,30,32,36,40,42,45,50,52,56,60,65,70,75,80,85,90,95,100,105,110,115,120,125,130,135,140 mm dia. **Black Square :** 40,50,55,63,75,80,85,90,95,100,105,110,115,120,125,140 mm Rounded Corner. Note : Supply condition- Rolled Material. Not available in Plate & Flat commodity.	Maintained Buffer Stock for Round Bar 25, 36, 45, 56, 75, 80 & 100 mm.
EN24	**Black Round :** 10,16,25,30,32,36,40,42,45,50,52,56,60,65,70,75,80,85,90,95,100,105,110,115,120,125,130,135,140 mm dia. **Black Square :** 40,50,55,63,75,80,85,90,95,100,105,110,115,120,125,140 mm Rounded Corner. Note : Supply condition- Rolled Material. Not available in Plate & Flat commodity.	Maintained Buffer Stock for Round Bar 20, 30, 40, 50, 60 & 80 mm.
SAE8620	**Black Round :** 10,16,25,30,32,36,40,42,45,50,52,56,60,65,70,75,80,85,90,95,100,105,110,115,120,125,130,135,140 mm dia. **Black Square :** 40,50,55,63,75,80,85,90,95,100,105,110,115,120,125,140 mm Rounded Corner. Note : Supply condition- Rolled Material. Not available in Plate & Flat commodity.	--
20MnCr5	**Black Round :** 10,16,25,30,32,36,40,42,45,50,52,56,60,65,70,75,80,85,90,95,100,105,110,115,120,125,130,135,140 mm dia. **Black Square :** 40,50,55,63,75,80,85,90,95,100,105,110,115,120,125,140 mm Rounded Corner. Note : Supply condition- Rolled Material. Not available in Plate & Flat commodity.	Maintained Buffer Stock for Round Bar 25, 36, 56, 75 & 100 mm.
C45 (Plate)	Thickness (in mm) 5, 6, 10 , 12, 16, 20, 22, 25, 28, 30, 32, 36, 40, 45, 50, 56, 63, 70, 75, 80, 90, 100, 110, 120, 130, 140 Standard width is 2500 x 6000mm.	--
En19	**Black Round :** 10,16,25,30,32,36,40,42,45,50,52,56,60,65,70,75,80,85,90,95,100,105,110,115,120,125,130,135,140 mm dia. **Black Square :** 40,50,55,63,75,80,85,90,95,100,105,110,115,120,125,140 mm Rounded Corner. Note : Supply condition- Rolled Material. Not available in Plate & Flat commodity.	Maintained BFRSTK for Round Bar 36, 45, 56, 75 & 100 mm.